

Patrick Kramer

(412) 214-3620 | pkramer@andrew.cmu.edu | patrick-kramer.github.io | github.com/patrick-kramer

EDUCATION

Carnegie Mellon University

PhD in Statistics | GPA: 3.99/4.00

Advisor: Edward H. Kennedy

Pittsburgh, PA

Expected: May 2029

Carnegie Mellon University

M.S. in Statistics | GPA: 4.05/4.00

Pittsburgh, PA

May 2026

Leipzig University

“Diplom” (German degree, equivalent to M.S.) in Mathematics with distinction | Class Rank: 1 of 65

Thesis: Regression Discontinuity Designs with Covariates (Advisor: Alexander Fuchs-Kreiss)

Leipzig, Germany

July 2024

Leipzig University

“Vordiplom” (German degree, equivalent to B.S.) in Mathematics | Class Rank: 1 of 65

Leipzig, Germany

May 2022

Leipzig University

B.S. in Computer Science | Class Rank: 1 of 252

Thesis: A Generalization of the Iteration Theorem for Recognizable Formal Power Series (Advisor: Andreas Maletti)

Leipzig, Germany

September 2019

RESEARCH INTERESTS

I am interested in Machine Learning, Causal Inference, and High-Dimensional Statistics, and particularly their intersection.

RESEARCH EXPERIENCE

Carnegie Mellon University

Mentors: Sivaraman Balakrishnan and Edward H. Kennedy

Project: Understanding optimal treatment policies through optimal transport

March 2026 - present

Pittsburgh, PA

Carnegie Mellon University

Mentor: Edward H. Kennedy

Project: Causal Inference with high-dimensional treatments

January 2025 - present

Pittsburgh, PA

Leipzig University

Mentor: Alexander Fuchs-Kreiss

Project: Estimating ATEs in regression discontinuity designs with covariates under minimal assumptions

August 2022 - October 2023

Leipzig, Germany

INVITED TALKS

Joint Statistical Meetings (JSM), Session for ASA Student Best Paper Award Winners

Causal Inference with High-Dimensional Treatments

Boston, MA

August 2026

INFORMS Annual Meeting

Causal Inference with High-Dimensional Treatments

Atlanta, GA

October 2025

PAPERS

Kramer, P., Kennedy, E. H., Oppen, I. M. **Causal Inference with High-Dimensional Treatments.** arXiv preprint arXiv:2602.21423, 2026. Honored with 2026 JSM Student Best Paper Award

Kramer, P., Fuchs-Kreiss, A. **Estimating Average Treatment Effects in Regression Discontinuity Designs with Covariates under Minimal Assumptions.** arXiv preprint arXiv:2310.08976, 2023. Submitted to Metrika, currently working on revisions.

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WORK EXPERIENCE

PricewaterhouseCoopers (PwC) Associate Finance Transformation	Leipzig, Germany February 2022 - June 2024
Check24 Front End App Developer for Android	Leipzig, Germany May 2020 - January 2022

TEACHING EXPERIENCE

Carnegie Mellon University Teaching Assistant - Introduction to Probability Modeling Teaching Assistant - Mathematical Statistics and Probability Theory	Pittsburgh, PA Spring 2025 Fall 2024
Leipzig University Teaching Assistant - Calculus for Computer Scientists Teaching Assistant - Computer Hardware Engineering Teaching Assistant - Modeling and Programming	Leipzig, Germany Fall 2018 Summer 2018 Summer 2018

HONORS AND AWARDS

ASA Student Best Paper Award (SLDS Section) American Statistical Association \$1000	January 2026
Top Student Award Top 1% of 350 students in 3 database courses Database Group, Leipzig University	December 2018

SCHOLARSHIPS

Deutschlandstipendium German Federal Ministry of Education and Research €3600	October 2021 - September 2022
Deutschlandstipendium German Federal Ministry of Education and Research €9000	October 2019 - March 2020

ACADEMIC SERVICE

Co-Organizer - Causal Inference Reading Group, Carnegie Mellon University	August 2025 - present
Member - PhD Professional Committee, Carnegie Mellon University	August 2024 - July 2025

SKILLS

- Programming Languages:** R, Python, Kotlin, Java, C, C++, SQL, JavaScript, MATLAB
- Technologies:** NumPy, Pandas, Scikit-Learn, TensorFlow, PyTorch, Git, Visual Studio Code, RStudio, LaTeX